GGG-F-451C
June 13, 1968
SUPERSEDING
Int. Fed. Spec. GGG-F-451B(GSA-FSS)
October 7, 1966 and
Fed. Spec. GGG-F-451A
April 5, 1961

FEDERAL SPECIFICATION

FLARING TOOL, TUBE HAND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand operated tools used for the purpose of flaring tubing.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification

1.2.1 Types, classes, and styles. Flaring tools shall be of the following types, classes, and styles, as specified (see 6.2).

Type I. Flaring, screw-actuated.
- Class 1. Roller cone.
  - Style A. Attached flare gage.
  - Style B. Without attached gage.
- Class 2. Swivel cone.
  - Style A. Hinged dies.
- Class 3. Combination, self-contained.

Type II. Flaring, impact-actuated.
- Class 1. Tube coupling.
- Class 2. Hand block.
  - Style B. Double flare.
- Class 3. Vise blocks.
- Class 4. Hinged blocks.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Box, Fiberboard.
WW-T-700/4c - Tube, aluminum alloy, drawn, seamless, 5052.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

FSC 5120

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-T-6845 - Tubing; Steel, Corrosion-Resisting, (18-8) Aircraft, Hydraulic System.
MIL-H-15424 - Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Military Standard Drawings:

MS20819 - Sleeve, Flared Tube Fitting.
MS 33583 - Tubing End - Standard Dimensions for Double Flare.
MS 33584 - Tubing End - Standard Dimensions for Flared.
MS 33656 - Fitting End Standard Dimensions for Flared Tube Connections and Gasket Seal.

Air Force-Navy Aeronautical Standard Drawings:

AN 815 - Union Flared Tube.
AN 818 - Nut, Coupling.
AN 929 - Cap Assembly - Pressure Seal Flared Tube Fitting.

Air Force-Navy Aeronautical Design Standard Drawings:

AND10106 - Tubing, Standard Sizes for Aluminum Alloy (52SO Round).

Air Force Drawing:

44A25450 - Sleeve Oxygen Coupling.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise inidcated, the issue in effect on date of invitation for bids shall apply.

Governmental:

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20406).

Nongovernmental:

Society of Automotive Engineers Standard:

SAE Handbook (Tube Fittings - Flared Tube).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York, New York 10017.)

3. REQUIREMENTS

3.1 The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of flaring tools otherwise conforming to this specification.

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3.2 Materials. The materials used shall be such as to produce flaring tools conforming to the requirements specified hereinafter. Material shall be free from all defects and imperfections that may adversely affect the serviceability of the finished product.

3.3 Finish and coatings. All parts shall be neatly finished and free from pits, nodules, burrs, cracks, or other detrimental defects. Surfaces subject to corrosion or rust shall have any one or a combination of the following coatings unless a choice is specified (see 6.2):

(a) Chromium plate
(b) Nickel plate
(c) Cadmium plate
(d) Zinc plate
(e) Hot phosphating
(f) Oxide coating

3.4 Design. Tubing tools of the latest design and essentially the standard product of the manufacturer will be acceptable provided they conform to this specification. All tools and components shall withstand the stresses applied during the maximum normal operations without sign of failure or permanent deformation in any respect. Components of one tool must be interchangeable with those of any other tool of the same type, class, and style made by the same manufacturer.

3.5 Marking for identification. Each tubing tool shall be marked by steel stamping or etching in a legible manner with the manufacturer's name or trademark, and the size (s) of tubing each tool will flare.

3.6 Threads. All threads conform to the applicable requirements of H28.

3.7 Performance and hardness tests. Tubing tools shall pass the performance concentricity and hardness tests specified in 4.4. When specified for Air Force purchases, they shall pass the additional tests specified in 4.4.4 for the tools indicated therein.

3.8 Repair parts. The successful bidders supplying flaring tools shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be completely interchangeable, and may be replaced or adjusted without requiring modification.

3.9 Carrying case. Unless otherwise specified carrying cases or boxes shall be of metal or other suitable material. The cases shall be of sturdy construction to withstand the test in 4.4.3. Both inside and outside surfaces of metal cases or boxes shall have a protective coat of paint.

3.10 Type I, class 1, style A, flaring, screw-actuated, roller cone, attached flare gage. Style A flaring tools shall be suitable for producing high-quality flares having a 74° included angle and conforming to Military Standard Drawing MS33584. The tool shall flare cold-drawn corrosion-resisting steel tubing of the type specified in Military Specification MIL-T-6845 having a yield point of 105,000 pounds per square inch, in sizes of 3/16, 1/4, 5/16, 3/8, 1/2, and 5/8-inch outside diameters. The attached flaring gage shall be easily operated, shall provide a positive means for flaring the tubing to the proper diameter, and shall be so secured that it will not become detached accidentally during normal use.

3.10.1 Design. The tool shall be designed primarily for maintenance use, and flares shall be made with a minimum application of torque capable of being exerted by an average mechanic working in a restricted position such as inside an aircraft wing. The tool shall be capable of withstanding the harder usage common in field service and shall operate adequately without distortion or misalignment. The tool shall consist essentially of a die-block assembly, a compression assembly, and a metal carrying case. When specified (see 6.2) a yoke-mounted precision-feed control reamer for removing burrs from the inner edge of tubing prior to flaring shall be furnished.

3.10.2 Die-block assembly. The assembly shall consist essentially of two parallel bars, the necessary number of die blocks, and a tube clamping device. The die blocks shall maintain proper alignment when mounted between the parallel bars. Substantial rivet-type pins or studs shall hold the bars to the clamping device at one end and to a die block at the other end. The clamping device shall have a swivel action to provide for quick removal of the tubing after flaring. The clamping screw shall have a hexagon extended washer head for tightening with a standard size wrench and the washer-head periphery shall be knurled for quick hand adjustment. The amount of potential squeeze between die blocks shall be sufficient to clamp tubing securely, but shall be held to a minimum to prevent excessive tube deformation, and in any case shall not exceed the following limits: 0.020 inch for 1/2 through 5/8 inch; 0.015 inch for 5/16 inch through 3/8 inch; and 0.012 inch for 1/4 inch and smaller tubes. The die blocks shall have a hardness of 47 to 57 in the Rockwell C scale or shall be casehardened to a minimum depth of 0.003 inch and shall show a surface hardness of 84 to 91 on the Rockwell 15-N scale. The point of the clamping screw shall be rounded and hardened or shall have a hardened steel ball inserted to resist scoring under hard usage. The true radius of the tube-gripping holes shall have a tolerance of plus or minus 0.002 inch,

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and the surface roughness of the holes shall not exceed a 100-microinch finish in order to minimize marring of the tubing during the flaring operation. The holes need not be countersunk as the flare is formed above the die blocks. The thickness of the blocks shall not exceed 1 inch so that flares may be made as close as possible to a bend in the tubing.

3.10.3 Compression assembly. The compression assembly shall provide the means of applying the pressure necessary for flaring, and shall consist essentially of a yoke, pressure sleeve and shaft assembly, and a roller flaring head.

3.10.3.1 Yoke. The yoke shall have a threaded hole to receive the pressure sleeve, and construction shall permit ready alignment over the tube-gripping holes in the die blocks. Means shall be provided for mounting a device to gage the setting of the tube in the die blocks and controlling the amount of flare.

3.10.3.1.1 A yoke extension shall be provided so that the yoke can be securely clamped in an ordinary bench vise.

3.10.3.2 Pressure sleeve and shaft assembly. The pressure sleeve shall have a thread no coarser than the American National extra-fine thread series to fit the yoke and shall have a suitably reamed hole to provide a standard running fit for the shaft. The shaft shall have a hexagon extended washer head for turning with a standard-size wrench and the periphery of the washer shall be knurled for quick hand adjustment. A driving lug or pin shall be securely fastened to the underside of the shaft head so as to drive against a slot or a lug integral with or securely fastened to the pressure sleeve. After the downward feed of the pressure sleeve has flared the tubing to the specified diameter, it is necessary that the roller flaring head rotate in the same plane sufficiently to true-up the flare, and means shall be provided to assure the success of this operation. The roller flaring head shall be securely fastened to the small end of the shaft in such manner as will preclude becoming detached during normal usage.

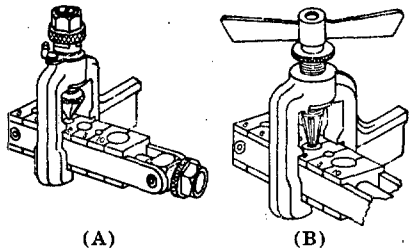
3.10.3.3 Roller flaring head. The flaring head shall have three or more rollers incorporated in such a manner that they rotate freely during the flaring operation. The rollers may be tapered or straight, but in either case, the surface shall have a maximum roughness of 16 microinches and the departure from straightness shall not exceed 0.002-inch per 1/2 inch length. The rollers shall have a hardness of 60 to 65 on the Rockwell C scale. The rollers shall not become detached during normal usage. The flaring head shall have a surface hardness of 80 to 86 on the Rockwell 15-N scale.

3.10.4 Carrying case. An adequate case (see 3.9) for holding all component parts of the tool shall be furnished. Whether a deburring reamer is furnished or not, the case dimensions shall not exceed a length of 11 inches, a width of 5 inches by a depth of 2 inches. The case shall have a hinged lid with either a positive or a firm friction catch.

3.10.5 Deburring reamer. If the deburring reamer is furnished (see 3.10.1 and 6.2), it shall be similar to the compression assembly specified under 3.10.3. The yoke shall be easily positioned over the tubing and shall be of the self-centering type. The sleeve shall have a thread no coarser than the American National fine-thread series, shall have a reamed hole to pilot the reamer and have a knurled head for control of feed. The tapered reamer shall have sufficient flutes and shall be of adequate size to deburr the range of tubing specified for the flaring tool. The shaft shall be a running fit in the threaded sleeve, shall have a handle securely attached on one end, and shall be integral with or securely attached to the tapered reamer. The handle shall be of adequate design to enable efficient deburring of the range of tubing specified for the flaring tool. The reamer shall have a hardness of 55 to 62 on the Rockwell C scale.

3.10.6 Weight. The weight of the complete tool with deburring reamer and carrying case shall not exceed 4 pounds.

3.10.7 Type I, class 1, style A flaring tools and deburring reamers shall be similar to figure 1, (A) and (B), respectively.



(A) (B)

FIGURE 1. Type I, class 1, style A flaring tool, screw-actuated, roller cone, attached flare gage.

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3.11 Type I, class 1, style B flaring tool, screw-actuated, roller cone, without attached gage. Style B flaring tools shall be in accordance with the requirements for style A tools specified in 3.10 except for the following differences:

(a) Tools shall flare fully annealed corrosion-resisting-steel tubing but not required to flare it in the cold-drawn condition.

(b) Capacity of tool shall be 3/4, 7/8, 1, and 1-1/4 inch outside diameter tubing.

(c) Not furnished with a precision-feed control deburring reamer.

(d) The thickness of the die blocks shall not exceed 1-1/2 inches.

(e) Not furnished with an attached flaring gage.

(f) Carrying case may exceed dimensions specified in 3.10.4 in order to adequately contain the tool.

(g) Maximum limit of potential squeeze shall be 0.035 inch for 3/4 inch and larger tubing.

3.11.1 Type I, class 1, style B flaring tools shall be similar to figure 2.

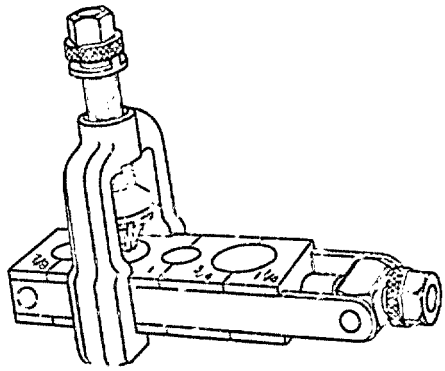


FIGURE 2. Type I, class 1, style B flaring tool, screw-actuated, roller cone without attached gage.

3.12 Type I, class 2, style A, flaring tool, screw-actuated, swivel cone, hinged dies. Style A flaring tools shall produce 90° included-angle flares on soft steel, copper, and aluminum tubing to fit standard S.A.E. tube fittings. They shall consist essentially of die bars and a compression assembly. When specified (see 6.2), they shall be furnished in boxes (see 3.9).

3.12.1 Double-lap flares. When specified (see 6.2), hardened adapter sets suitable for producing double-lap flares on 3/16, 1/4, 5/16, 3/8, 1/2, 5/8 and 3/4 inch tubing to fit standard S.A.E. tube fittings shall be furnished. The hardness shall comply with requirements specified in 3.12.3.3.

3.12.2 Die bars. The bars shall be hinged at one end to permit easy removal of the tubing after flaring. Wing nuts or other suitable means shall be provided to securely clamp the tubing during the flaring operation. The tube gripping holes shall be serrated or other means provided to prevent the tubing from slipping during the flaring operation and the holes shall be properly countersunk. The amount of potential squeeze between die blocks shall be sufficient to clamp tubing securely, but shall be held to a minimum to prevent excessive tube deformation, and in any case shall not exceed the following limits: 0.040 inch for 5/8 inch and larger; 0.035 inch for 3/8 through 1/2 inch; and 0.030 inch for 5/16 inch and smaller tubes. The bars shall have a hardness of 47 to 57 on the Rockwell C scale or be case-hardened to a minimum depth of 0.003 inch with a surface hardness of 84 to 91 on the Rockwell 15-N scale. In order to cover the range of tubing specified, tool size J (see table I) may be furnished with two pairs of die bars.

3.12.3 Compression assembly. The compression assembly shall provide the means of applying the necessary pressure for the flaring operation and shall consist essentially of a yoke, pressure screw and handle, and a flaring cone.

3.12.3.1 Yoke. The yoke shall have a threaded hole to receive the pressure screw, and the yoke construction shall permit ready alignment over the tube-gripping holes in the hinged bars, so that finished flares will meet the concentricity requirements specified in 4.4.1.2. Both hinged bars shall be supported the same height within 0.003 inch during the flaring operation.

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3.12.3.2 Pressure screw and handle. The pressure screw shall provide for the mounting of a swiveling flaring cone which does not rotate when the screw is turned during the flaring operation. A T-type sliding handle of adequate length to produce the required torque shall be provided on the opposite end. The handle shall not become detached accidentally from the pressure screw during normal usage. When specified (see 6.2), the screw shall have an extended additional head 1/4 inch square which permits use of a 1/4 inch ratchet wrench for working in close quarters. All pressure screws having the extended 1/4 inch square head shall have the T-type sliding handle also.

3.12.3.3 Flaring cone. The flaring cone shall be of the swiveling type and shall be mounted on the pressure screw in such manner that it will revolve freely but will not become detached under hard usage. The conical face shall have a maximum surface roughness of 32 microinches. The cone shall have a hardness of 51 to 60 on the Rockwell C scale or shall be case-hardened to a minimum depth of 0.020 inch and shall show a hardness of 86 to 91 on the Rockwell 15-N scale.

3.12.4 Type I, class 2, style A flaring tools shall be similar to figure 3 and shall conform to table I for the tool letter specified (see 6.2).

Table I. Type I, class 2, style A flaring tool, screw-actuated, swivel cone, hinged dies

Tool letter	Range of tool, outside diameter of tubing in inches											
A	----	----	----	----	----	----	----	----	3/4	7/8	1	----
B	----	----	----	----	----	----	----	5/8	---	7/8	---	1-1/8
F	----	3/16	1/4	5/16	3/8	----	1/2	5/8	---			
J	1/8	3/16	1/4	5/16	3/8	7/16	1/2	5/8	3/4			

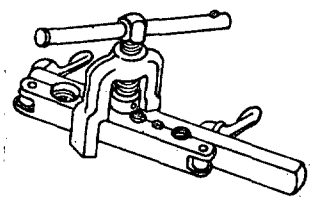


FIGURE 3. Type I, class 2, style A flaring tool, screw-actuated, swivel cone, hinged dies.

3.13 Type I, class 3, flaring tool, screw-actuated combination, self-contained. Class 3 flaring tools shall be suitable for producing high-quality flares having a 74° included angle. The tool shall be capable of single flaring of all grades of aircraft tubing including one-quarter hard corrosion resisting steel, sizes 1/8 inch through 3/4 inch, in accordance with Military Standard Drawing MS 33584. When specified (see 6.2) the tool shall produce 90° flares on soft copper and aluminum tubing in outside diameters 1/8 inch through 3/4 inch to fit standard S.A.E. fittings. The tool shall also be capable of double-lap flaring of mild aluminum tubing, sizes 3/16 inch through 3/4 inch. The double-lap flaring of tubing sizes 3/16 inch through 3/8 inch shall be in accordance with Military Standard Drawing MS 33583.

3.13.1 Design. The tool shall be designed primarily for maintenance use and flares shall be made with a minimum application of torque capable of being exerted by an average mechanic working in a restricted position such as inside an aircraft wing. The tool shall be capable of withstanding the harder usage common in field service and shall operate adequately without permanent distortion or misalignment. The tool shall incorporate tube-setting stops or other positive means for flaring the tubing to the proper diameter. The first operation punches for double-lap flaring and the finish flaring cone shall be self-contained in the tool. The tool shall have a vise-grip attachment so that it can be used as a bench tool.

3.13.1.1 Material and hardness. The flaring cone and first operation punch shall be made from high grade carbon bar steel (SAE 1020 minimum). The cone and punch shall be case hardened to a depth of at least 0.040 inch and have a hardness of at least 89 on the Rockwell 15-N scale.

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3.13.2 Die (tube clamping) blocks. Die blocks shall be made of bar steel or powdered moulded steel. When made of bar steel, die blocks shall have a minimum hardness of Rockwell "C" 47, or shall be case hardened to a minimum depth of 0.015 inch and show a minimum surface hardness of 84 on the Rockwell 15-N scale. When made of powdered moulded steel, die blocks shall be carbonitrated and shall have a minimum hardness of 65 on the Rockwell 15-N scale.

3.13.3 Carrying case. An adequate case (see 3.9) for the tool with compartments to accommodate all component parts shall be furnished. The case shall have a hinged lid with a positive catch.

3.13.4 Weight. The weight of the tool including any one set of die blocks and the vise grip attachment shall not exceed 8 pounds. The weight of the tool complete with all components and the carrying case shall not exceed 10-1/2 pounds.

3.13.5 Operating instructions. Each tool shall have a set of operating instructions enclosed within the carrying case.

3.13.6 Type I, class 3 flaring tools shall be similar to figure 4.

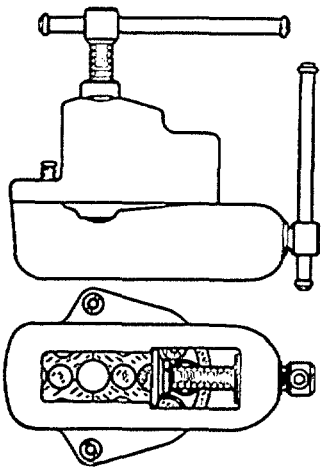


FIGURE 4. Type I, class 3 flaring tool, screw-actuated, combination self-contained.

3.14 Type II, class 1 flaring tool, impact-actuated, tube coupling. Class 1 flaring tools shall consist essentially of a guide sleeve adapter and a flaring punch. The 74° tool shall flare tubes conforming to flare dimensions of Military Standard Drawing MS 33584.

3.14.1 Guide sleeve adapter. The adapter shall be either threaded externally at one end for engaging an AN 818, or equal, (see Air Force-Navy Aeronautical Standard Drawing (AN818) triple-type tube coupling nut, or shall be recessed or counterbored for receiving an MS 20819, or equal, (see Military Standard Drawing MS 20819) sleeve through a "side-access" opening. The "side-access" type of adapter shall have a spring clip or other suitable means of retaining the sleeve after it has been inserted in the adapter. Both types of adapters shall have a reamed hole to provide a free sliding fit for the flaring punch. The reamed hole shall be concentric with the locating threads or the locating recess within 0.005 inch total indicator reading.

3.14.2 Flaring punch. The punch shall be cone-shaped at one end, with the striking head integral with or permanently fastened at the opposite end. The punch shall show a hardness of 37 to 52 on the Rockwell "C" scale. The striking head shall resist mushrooming or chipping and the cone shall not score during hard usage. The conical face shall have a maximum surface roughness of 32 microinches and the departure from straightness shall not exceed 0.002 inch per 1 inch length. The cone and the body shall be concentric within 0.005 inch total indicator reading.

3.14.3 Use data. Class 1 flaring tools shall utilize the triple-type sleeve in the coupling unit as the female die in the flaring operation, and shall be capable of flaring tubing which has a bend immediately adjacent to its end.

3.14.4 Type II, class 1 flaring tools shall conform to table II for the tool number specified (see 6.2) and shall be similar to figure 5 (A) or (B).

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TABLE II. Type II, class 1 flaring tool, impact-actuated, tube coupling

Tool number	Applicable tube fitting	Included angle of flaring punch ± 1/2°	Tubing size nominal O. D.	Material and maximum wall thickness of tubing to be flared		
				Soft copper or aluminum	Steel fully annealed	Corrosion-resisting steel fully annealed
		Degrees	Inches	Inch	Inch	Inch
4A	AN	74	1/4	0.035	0.028	0.022
5A	AN	74	5/16	.035	.028	.022
6A	AN	74	3/8	.042	.032	.028
8A	AN	74	1/2	.049	.035	.032
10A	AN	74	5/8	.049	.035	.032
12A	AN	74	3/4	.065	.049	.035
16A	AN	74	1	.065	.049	.035
20A	AN	74	1-1/4	.065	.049	----
24A	AN	74	1-1/2	.065	.049	----

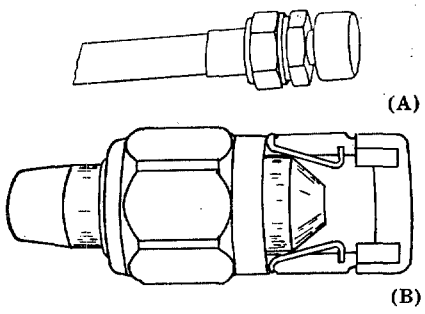


FIGURE 5. Type II, class 1 flaring tool, impact-actuated, tube coupling.

3.15 Type II, class 2 flaring tool, impact-actuated, hand block. Class 2 flaring tools shall be suitable for holding in the hand while performing the flaring operation.

3.15.1 Style B, double flare. Style B flaring tools shall be suitable for double-lap 74° flares in accordance with Military Standard Drawing MS 33583, aluminum alloy tubing conforming to Federal Specification WW-T-700/4C having a nominal outside diameter of 5/16 inch and a wall thickness of 0.032 inch, or a wall thickness of 0.035 inch in accordance with Air-Force-Aeronautical Design Standard Drawing AND 10106.

3.15.1.1 Design. The flaring tool shall have positive stops to prevent over flaring of the tubing and all parts subject to wear shall be heat-treated to have a hardness of 37 to 52 on the Rockwell "C" scale or shall be case-hardened to a minimum depth 0.015 inch and have a surface hardness of 78 to 86 on the Rockwell 15-N scale. Each flaring tool shall be furnished with a small, light weight deburring tool capable of removing all burrs from the inner and outer edges of the tubing end.

3.15.1.2 Carrying case. An adequate steel case for holding all component parts of the tool shall be furnished with each flaring tool. It shall resist the entrance of water in rainy weather, shall be equipped with a piano type continuous hinge, and shall have a hasp and staple. A flush type metal handle shall be located properly on the top side of the case. The tool case shall be of sturdy construction to withstand the test in 4.4.3. Both the interior and exterior of the case shall have a protective coat of paint.

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3.15.1.3 Dimensions. The maximum outside dimensions of the carrying case shall not exceed 7 by 5 by 4 inches. The flaring tool, complete with all adapters, shall not exceed 5 inches in length and 3-1/2 inches in diameter.

3.15.1.4 Weight. The flaring tool, complete with all adapters and carrying case, shall weigh not more than 4-1/2 pounds.

3.15.1.5 Type II, class 2, style B flaring tool shall be similar to figure 6.

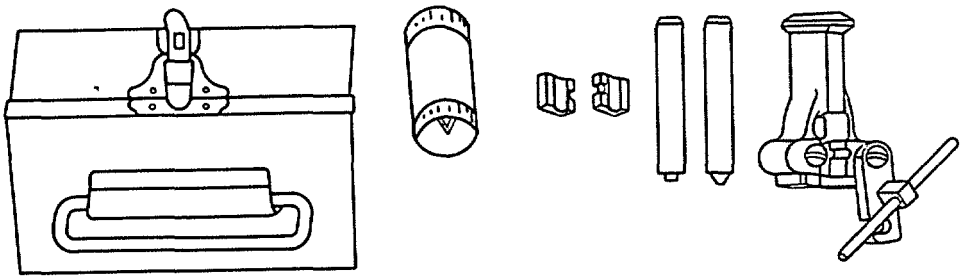


FIGURE 6. Type II, class 2, style B double-flaring tool, impact-actuated, hand block.

3.16 Type II, class 3 flaring tool, impact-actuated, vise blocks. Class 3 flaring tools shall consist essentially of die blocks and a flaring punch. The tool shall flare tubes conforming to the flare dimensions of Military Standard Drawing MS 33584.

3.16.1 Die blocks. The blocks shall be fitted with two or more dowel pins to insure positive alignment of the countersunk hole during the flaring operation. The dowel pins shall fit securely in one die block and shall be a sliding fit in the other. The blocks are designed for use in a standard bench vise which provides the necessary clamping action to hold the tubing. The amount of potential squeeze between die blocks shall be sufficient to clamp tubing securely, but shall be held to a minimum to prevent excessive tube deformation, and in any case shall not exceed 0.035 inch for 1-1/2 inch diameter and larger tubing and 0.025 inch for tubing smaller than 1-1/2 inch. The true diameters of the tube gripping section of the die blocks shall have a tolerance of minus 0.000, plus 0.003 inch for diameters 1-1/2 inches or under, and minus 0.000, plus 0.006 inch for diameters over 1-1/2 inches. The surfaces roughness of the tube gripping section shall not exceed a 100 microinch finish. The countersink shall be concentric with the hole within 0.005 inch total indicator reading. The blocks shall be case hardened to a minimum depth of 0.015 inch and show a surface hardness of 78 to 86 on the Rockwell 15-N scale.

3.16.2 Flaring punch. The punch shall conform to 3.13.2, except that a pilot shall be provided on the cone-shaped end and shall be a sliding fit in the applicable size tube to insure proper centering, and shall be concentric with the cone within 0.005 inch total indicator reading. The minimum length of the pilot shall be the same dimension as the inside diameter of the applicable tube, When ordering class 3 flaring tools or extra punches, the wall thickness of the tubing should be specified in the contract or order, in order that punches be furnished with the correct pilot diameter (see 6.2).

3.16.3 Type II, class 3 flaring tools shall conform to table III for the tool number specified (see 6.2) and shall be similar to figure 7.

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TABLE III. Type II, class 3 flaring tool, impact-actuated, vise blocks

Tool number	Applicable tube fitting	Included angle of flaring punch plus or minus 1/2°	Tubing size nominal O. D.	Maximum wall thickness of tubing to be flared for soft copper or aluminum, and steel or corrosion-resisting steel fully annealed.
		Degree	Inches	Inch
4	AN	74	1/4	0.065
6	AN	74	3/8	.083
16	AN	74	1	.120
32	AN	74	2	.109

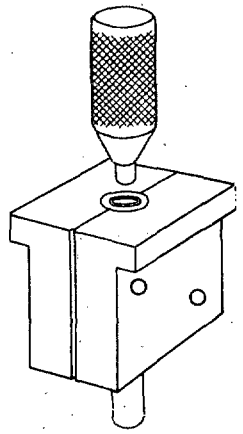


FIGURE 7. Type II, class 3 flaring tool, impact-actuated, vise blocks.

3.17 Type II, class 4, flaring tool, impact-actuated, hinged blocks. Class 4 flaring tools shall consist essentially of die bars and a yoke assembly. The tool shall flare tubes conforming to flare dimensions of Military Standard Drawing MS 33584.

3.17.1 Die bars. The die bars shall have seven countersunk tube-clamping holes and shall be hinged at one end in such a manner as to insure positive alignment of each size hole and easy removal of the tubing after flaring. Provisions for centralizing each tubing hole with the flaring punch in the yoke assembly shall be incorporated in one or both of the die bars. The countersinks shall be concentric with the holes within 0.005 inch total indicator reading. The amount of potential squeese between die bars shall be sufficient to clamp tubing securely, but shall be held to a minimum to prevent excessive tube deformation, and in any case shall not exceed the following limits: 0.040 inch for 5/8 inch and larger; 0.035 inch for 3/8 through 1/2 inch; and 0.020 inch for 5/16 inch and smaller tubes. The die bars shall be case hardened to a minimum depth of 0.015 inch and show a surface hardness of 78 to 86 on the Rockwell 15-N scale.

3.17.2 Yoke assembly. The yoke assembly shall consist essentially of a yoke, flaring punch, and a clamping screw.

3.17.2.1 Yoke. The yoke shall have a reamed hole perpendicular to the countersunk side of the die bars and provide a sliding fit for the flaring punch. The side of the yoke shall have a threaded hole to receive the clamping screw. Provisions shall be made for a spring catch or other means to hold the flaring punch in a raised position while sliding the yoke from one location to another. The yoke shall have a feature for self-centering over each hole when the clamping screw has been tightened ready for flaring.

3.17.2.2 Flaring punch. The punch shall conform to 3.14.2.

3.17.2.3 Clamping screw. The clamping screw shall be provided with a T-type sliding handle of adequate length to produce the proper torque when hand tightened by the average artisan. The sliding handle shall not become detached accidentally from the screw under normal usage. The point of the screw shall be oval-shaped and hardened to Rockwell "C" 47 to 52, or to 78 to 86 on the Rockwell 15-N scale when case hardened to prevent scoring when tightened against the die bars.

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3.17.3 Type II, class 4 flaring tools shall conform to table IV and shall be similar to figure 8.

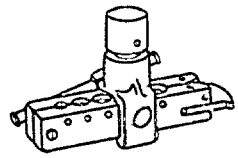


FIGURE 8. Type II, class 4 flaring tool, impact-actuated, hinged blocks.

TABLE IV. Type II, class 4 flaring tool, impact-actuated, hinged blocks

Applicable tube fitting	Included angle of flaring punch plus or minus 1/2°	Tubing size nominal O. D.	Material and maximum wall thickness of tubing to be flared		
			Soft copper or aluminum	Steel fully annealed	Corrosion-resisting steel fully annealed
	Degrees	Inch	Inch	Inch	Inch
AN	74	1/8	0.035	0.035	0.028
AN	74	3/16	.035	.035	.028
AN	74	1/4	.065	.049	.049
AN	74	5/16	.065	.049	.049
AN	74	3/8	.065	.065	.049
AN	74	1/2	.083	.065	.065
AN	74	5/8	.095	.072	.065

3.18 Workmanship. Workmanship shall be first class in every respect. Edges and corners shall be rounded in conformance with the best commercial standards. The tools shall have no burrs, fins, sharp projections, excessive surface roughness, or any other imperfections which may impair their durability and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table V.

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TABLE V. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level	AQL Expressed in terms of	Reference
Visual examination	1 ea.	II	2.5 Major 4.0 Total	Defects per hundred units	4.3.1
Dimensional and weight examination	1 ea.	S-3	2.5	Defects per hundred units	4.3.2
Testing	1 ea.	S-3	2.5	Defects per hundred units	4.4.1 4.4.4.2
Testing of case	1 ea.	S-1	1.0	Percent defective	4.4.3

4.3 Examination

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanhsip, and marking. Defects are listed in table VI.

TABLE VI. Classification of defects

Defects
Critical
None defined
Major
Type, class, style, size, tool number, diameter of pilot and angle of flare not as specified.
Tool incomplete.
Materials defective; evidence of pits, nodules; burrs, or cracks.
Any component not aligned with mating part.
Clamping device not securely held in place or fails to swivel easily (type I, class 1, styles A and B only).
Clamping device not securely held in place, clamping device not easily swiveled, sliding handle not of sufficient length, or sliding handle not prevented from becoming detached (type I, class 3, type II, class 2, style B).
Clamping screw head or shaft head defective, not knurled or not as specified (type I, class 1, styles A and B).
Clamping screw defective, point not rounded, sliding handle not of sufficient length, or sliding handle not prevented from becoming detached (type II, class 4 only).
True diameter of tube-gripping holes not within the specified tolerance (type I, class 1, styles A and B).
True diameter of tube-gripping holes not within the specified tolerance or holes not countersunk (type II, class 4).
Tube-gripping holes not countersunk, or not provided with a means of preventing the tubes from slipping (type I, class 2, style A, type II, class 3).
Thickness of die blocks exceeds the specified maximum (type I, class 1, styles A and B).
Pressure screw defective, sliding handle not prevented from becoming detached, handle not of sufficient length, or extended square head (if required) missing or not as specified (type I, class 2, style A, type I, class 3).
Die bars not supported the same height (by the yoke) within the maximum allowable (type I, class 2, style A only).
Wing nuts fail to clamp bars securely in place, or fail to permit easy removal of the tubing (type I, class 2, style A only).
Gage missing or defective (type I, class 1, style A only).

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TABLE VI. Classification of defects (con't)

Defects
Driving lugs on underside of shaft head and on the pressure sleeve not securely fastened, or lugs fail to drive against each other (type I, class 1, styles A and B only).
Yoke assembly fails to permit ready alignment over the tube-gripping holes (type I, class 1, styles A and B, type I, class 2, style A).
Deburring reamer (if required) missing, defective, yoke not easily positioned, reamer defective, handle not secure, or tool not as specified (type I, class 1, styles A and B).
Not provided with a vise grip attachment (type I, class 3).
Not provided with positive stops to prevent over flaring (type II, class 2, style B).
Not provided with a deburring tool, tool defective, or not capable of removing burrs from both inner and outer surfaces (type II, class 2, style B).
Failure to provide a means for allowing the flaring head to rotate in the same plane (for truing the flare) (type I, class 1, styles A and B).
Flaring head not securely fastened to shaft, rollers not free rotating, or rollers not mounted so as to prevent them from becoming detached during normal usage (type I, class 1, styles A and B).
Flaring cone defective not free to revolve, or not properly mounted on end of pressure screw (type I, class 2, style A).
Flaring cone defective, not securely attached to pressure screw, not finished as required, or cone not aligned with die blocks (type I, class 3).
Not provided with a means of aligning the flaring punch over each tube-gripping hole (type II, class 4).
Not provided with tube setting stops or other means for flaring the tubing to the proper diameter (type I, class 3).
Punch defective, chipped or striking head not permanently fastened (type II, class 1, type II, class 3, type II, class 4).
Punches defective, chipped, not finished as required, or striking heads not beveled (type II, class 2, style B).
Flaring punch not provided with a spring catch to hold it in the raised position (type II, class 4).
Threads on adapter defective, torn, stripped, crossed, or not as specified (type II, class 1).
Hole in adapter fails to provide a free sliding fit for the punch (type II, class 1).
Triple type sleeve (die) defective or not aligned (type II, class 1).
Length of pilot not as required (type II, class 3).
Minor
Carrying case defective, fails to provide compartments for all components, hardware (handle, hinges, and catch) defective or inoperable, or painting defective or incomplete (type I, class 3, type II, class 2, style B).
Carrying case outside dimensions exceed the specified maximum values (type II, class 2, style B).
Flaring tool dimensions exceed the specified maximum values (type II, class 2, style B).
Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanently marked.

4.3.2 Dimensional and weight examination. Each sample unit shall be examined for any nonconformance with dimensional and weight requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 through 4.4.3. Air Force purchases shall have additional tests in accordance with 4.4.4.1 and 4.4.4.2.

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4.4.1 Performance test. Each of the tools selected in accordance with 4.2 shall flare one end of at least two pieces of each size of tubing which the tool accommodates. Tubing of the material and having a wall thickness equal to or greater than that shown in table VII opposite the applicable type, class and style of tool shall be used for this test. When a deburring reamer is furnished (see 3.10.5), it shall be used to deburr the tubing used for testing the parent tool. The tubing end to be flared shall be cut off clean, square, and the sharp edges smoothly deburred. All flares shall be smooth and free from cracks and pit marks. During and after the performance test, there shall be no evidence of scoring or scratching of of the cone or rollers; metal adhering to the cone or rollers; failure of any component to function properly; difficulty in removal or insertion of replaceable parts; or permanent deformation or damage of components in any respect. In addition, the inspection of all tubes flared by type I, class 1, styles A and B; type II, classes 1, 2, 3 and 4 tools shall show no cutting, scratching, or marring of the tubing surfaces which may under vibration and strain cause the tubing to fail in service. Flare concentricity shall be determined by checking at the midway point on the flare and at point 1/4 inch below the flare on the tubing. Concentricity shall be within the allowable tolerances specified for the applicable types, classes and styles.

Table VII. Material and nominal dimensions of tubing used in the performance test for flaring

Operation performed	Tool designation			Outside diameters of tubing in inches														Material
	Type	Class	Style	1/8	3/16	1/4	5/16	3/8	7/16	1/2	5/8	3/4	7/8	1	1-1/8	1-1/4	1-1/2	
				Wall thickness (inches)														
Single flare	I	1	A	---	0.032	0.032	0.032	0.032	---	0.035	0.035	---	---	0.065	---	0.083	---	Cold drawn corrosion-resisting steel
Single flare	I	1	B	---	---	---	---	---	---	---	---	0.065	0.065	0.065	---	0.083	---	Annealed corrosion-resisting steel
Single flare	I	2	A	0.020	.028	.035	.035	.035	.035	.035	.049	.049	.049	.058	0.058	---	---	Steel annealed
Double-lap flare	I	2	A	---	.028	.028	.032	.032	---	.032	.032	.032	---	---	---	---	---	Steel annealed
Single flare	I	3	---	---	.028	.028	.032	.032	---	.035	.035	---	---	---	---	---	---	Cold drawn corrosion-resisting steel
Double-lap flare	I	3	---	---	---	.028	.028	.032	---	---	---	---	---	---	---	---	---	Mild aluminum
Single flare	II	1	---	---	---	.028	.028	.032	---	.035	.035	.049	---	.049	---	.049	.049	Steel annealed
Double-lap flare	II	2	B	---	---	---	.028	---	---	---	.095	---	---	.120	---	---	---	Annealed aluminum alloy
Single flare	II	3	B	---	---	---	.028	---	---	---	.095	---	---	.120	---	---	---	Steel annealed
Single flare	II	4	---	---	---	.049	.049	.065	---	.065	.072	---	---	---	---	---	---	Steel annealed

4.4.1.1 Type I, class 1, styles A and B; class 3; type II, classes 1, 2, 3 and 4. The 74° single-lap flares produced by these tools shall conform to Military Standard Drawing MS 33584, and in addition shall be concentric with the outside diameter of tubing within 0.005 inch total indicator reading on tubing below 3/4 inch diameter and within 0.008 inch on tubing 3/4 inch diameter and over.

4.4.1.1.1 Type I, class 3; type II, class 2, style B. The 74° double-lap flares produced by these tools shall conform to Military Standard Drawing MS 33583.

4.4.1.2 Type I, class 2, style A. The 90° single flares produced by these tools shall be in accordance with S.A.E. Handbook (Tube Fittings - Flare Tube), except they shall be concentric with the outside diameter of the tubing within 0.012 inch total indicator reading, and the included angle inside the flare shall be between 88° and 92°.

4.4.1.3 Type I, class 2, style A. The 90° double-lap flares produced by these tools shall meet the requirements specified for 90° single-lap flares (see 4.4.1.2).

4.4.1.4 Type II, classes 1 and 3. The 74° flares produced by these tools shall be concentric with the outisde diameter of the tubing within 0.008 inch total indicator reading.

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4.4.2 Hardness test. Each of the sample tools selected in accordance with 4.2 shall be tested by method 243.1 of Federal Test Method Standard No. 151 for the applicable hardness requirements of section 3.

4.4.3 Drop test. Each of the sample cases (complete with tools) selected in accordance with 4.2 shall be dropped to land with a flat impact on its base five times from a height of 4 feet onto a concrete floor or slab. The case and methods of holding the contents in place shall be such that neither the case nor contents will be damaged to a degree detrimental to use or serviceability, by five free fall drops from a height of 4 feet onto a concrete floor or slab.

4.4.4 Tests. When specified for Air Force purchases, a hydraulic pressure test and a gas pressure test shall be conducted in addition to the tests specified in 4.4.1 through 4.4.3 for the types, classes, and styles indicated below.

4.4.4.1 Hydraulic pressure test. Each sample 1/2 inch size of type I, class 1, style A; type I, class 3 and one inch size of type II, class 3 flaring tool selected in accordance with 4.2 shall be tested as follows: Flare 1/2 inch O. D. by 0.049 inch wall thickness and 1.0 inch O. D. by 0.120 inch wall thickness annealed corrosion-resisting steel hydraulic tubing on both ends. Assemble one sample of tubing for each tool selected with coupling nuts (Drawing AN 818) and sleeves (MS 20819) to a union (Drawing AN 815) with a cap assembly (Drawing AN 929) attached. Connect tubing assembly together using a wrench torque of 450-inch-pounds for 1/2 inch size, and 1620 inch-pounds for 1.0 inch size. The tubing assembly shall then be connected to a hydraulic pressure source (oil or clear water) and subjected to a pressure of 6,000 pounds per square inch. There shall be no evidence of any leakage of the flared tube connections.

4.4.4.2 Gas pressure test. Each sample of type II, class 2, style B flaring tool selected in accordance with 4.2 shall be tested as follows: Flare on both ends, one piece of aluminum alloy tubing conforming to Federal Specification WW-T-700/4C having an outside diameter of 5/16 inch and a wall thickness of 0.035 inch. Assemble one piece of tubing (flared by each tool) with coupling nuts conforming to Air-Force-Navy Aeronautical Standard Drawing AN 818 and with coupling sleeves conforming to Air Force Drawing 44A25450. The tube assemblies shall be connected together, using a wrench torque of 100 inch-pounds and no thread lubrication, by means of tube fittings with ends in accordance with Military Standards Drawing MS 33656. The tubing assembly shall then be connected to a gas pressure source, immersed in clean water, and subjected to a gas pressure of 1000 pounds per square inch. There shall be no leakage of the flared tube connections as evidenced by any bubble formation in the water at the connections.

4.4.5 Possible test failures. Possible test failures are defined as follows:

(a) Hardness:

(1) Hardness of die blocks, flaring head, or rollers, not within the specified values on the Rockwell C or 15-N scale (type I, class 1, styles A and B, type I, class 3).

(2) Hardness of cone or die bars (dies or die blocks) not within the specified value on the Rockwell C or 15-N scale (type I, class 2, style A).

(3) Hardness of die blocks (cast iron) less than the minimum Brinell value specified, hardness of die blocks, punch (steel), or parts subject to wear not within the specified values on the Rockwell C or 15-N scales (types II, class 2, style B).

(4) Hardness of punch die bars or die blocks not within the specified values on Rockwell C or 15-N scale (type II, classes 1, 3 and 4).

(b) Finish:

Surface roughness of tube-gripping holes, flaring head rollers, cone, conical face of the punch (except type II, class 2, style B).

(c) Straightness of rollers:

Rollers depart from straightness by more than the maximum allowable (type I, class 1, styles A and B).

(d) Straightness:

Cone face departure from straightness exceeds the maximum allowable (type II, classes 1, 3 and 4).

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(e) Concentricity:

Concentricity of reamed holes, tube-clamping holes, and countersinks, or concentricity of cone face with the body of the punch not within the specified total indicator readings (type II, classes 1, 3 and 4).

(f) Weight:

Weight of complete tool with carrying case exceeds the maximum allowable (type I, class 3, type II, class 2, style B).

(g) Bar separation:

Separation between die bars exceeds the specified maximum when standard drill rod is clamped between them (type II, class 4).

(h) Block separation:

Separation between mating die blocks not within the specified maximum and minimum values when standard drill rods are clamped between them (type I, class 1, styles A and B, type I, class 2, style A, type II, classes 3 and 4).

(i) Drop test:

Evidence of fracture or damage to carrying case or its contents.

(j) Performance testing:

(1) Failure of tool to flare the specified tubes, or excessive torque required to flare tubes, or evidence of metal adhering to rollers after the test. Flare produced by tool fails to conform with specified requirements, evidence of cracks or scratches in the tubing, or outside diameter of flare not concentric with outside diameter of tube within the specified total indicator reading. Deburring tool (if required) fails to perform its required function die blocks distort the tubing more than the maximum allowable (type I, class 1, styles A and B, type I, class 2, style A).

(2) Failure of tool to flare the specified size tube, striking head mushrooms or chips, or cone is scored during the test. Flare produced by the tool fails to conform with specified requirements, evidence of cracks or pits in the tubing, or outside diameter of flare not concentric with outside diameter of tube within the specified total indicator reading (type II, classes 1, 3 and 4).

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A. The flaring tools shall be preserved and packaged in accordance with MIL-H-15424 as specified for level A, for tool, beading, and flaring (tubing).

5.1.2 Level C. The flaring tools shall be preserved and packaged in a manner to prevent deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. The flaring tools shall be packed in accordance with MIL-H-15424 as specified for level A, for tool, beading, and flaring (tubing). When specified (see 6.2), containers shall conform to overseas type of PPP-B-601. The plywood shall be water repellant treated in accordance with the box specification.

5.2.2 Level B. The flaring tools shall be packed in accordance with MIL-H-15424 as specified for level B, for tool beading, and flaring (tubing). When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with the box specification.

5.2.3 Level C. The flaring tools shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I, class 1, style A and type I, class 3 flaring tools are used for producing high-quality flares on corrosion-resisting steel tubing in both cold drawn and annealed state for aircraft hydraulic fuel, oil, and air lines. Type I, class 3 flaring tools are also used for producing high quality double lap flares on mild aluminum tubing used in aircraft.

6.1.2 Type I, class 1, style B flaring tools are used for producing high quality flares on steel and annealed corrosion-resisting steel tubing for hydraulic and other equipment requiring strong flares on various weights of tubing.

6.1.3 Type I, class 2, style A flaring tools are used for producing single and double-lap flares of a general purpose quality on soft copper and steel tubing for automotive fuel, oil, and water lines, and other general purpose equipment.

6.1.4 Type II, classes 1 and 4 flaring tools are used for producing good-quality flares on copper, aluminum, and soft steel tubing for hydraulic, steam, refrigeration, and instrumentation lines.

6.1.5 Type II, class 3 flaring tools are used for producing good-quality flares on steel, copper, and aluminum tubing having a heavy wall thickness for use in hydraulic systems, etc.

6.1.6 Type II, class 2, style B flaring tools are used for double-lap flaring of annealed aluminum alloy tubing used in oxygen systems for aircraft.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class and style if applicable, of tool required (see 1.2).
(c) Finish, if a choice is required (see 3.3).
(d) Tool number or letter, as applicable, of tool required (see tables I, II, and III).
(e) Precision feed control deburring reamer, if required for type I, class 1, style A flaring tool (see 3.10.1 and 3.10.5).
(f) Boxes, if required for type I, class 2, style A flaring tools (see 3.12).
(g) Adapter set for double lap flares, if required for type I, class 2, style A flaring tool (see 3.12.1).
(h) Extended head 1/4 inch square, if required for type I, class 2, style A (see 3.12.3.2).
(i) Flares having 90° included angle, if required for type I, class 3 flaring tool (see 3.13).
(j) Specify wall thickness of tubing that is to be flared in addition to the tool size number, for type II, class 3 flaring tools (see 3.16.2).
(k) Level of preservation, packaging and packing required (see 5.1 and 5.2).
(l) Marking, if different (see 5.3).

6.2.1 When ordering repair parts, specify the following:

(a) The manufacturer's name, or trademark of the tool which the parts are for.
(b) Type, class, and style of tool, if applicable.
(c) Tool number, if applicable (see tables I, II and III).
(d) Included angle of flare in degrees and wall thickness of tubing, if applicable.
(e) The name or description of the part, and size if applicable.
(f) Quantity of each part required.

6.3 For civil agency procurements. When level B packaging is required by civil agency procurements, the level A requirements shall apply.

6.4 Supersession data. The classification of flaring tools in this specification, corresponding to those covered in Federal Specification GGG-F-451A are as follows:

GGG-F-451A	GGG-F-451C
Type I. Flaring, screw-actuated	Type I
Class 1 - Roller cone	Class 1
Style A - Attached flare gage	Style A
Style B - Without attached gage	Style B
Class 2 - Swivel cone	Class 2
Style A - Hinged dies	Style A
Style B - Block dies	Deleted
Style C - Dowelled dies	Deleted
Style D - Indexing dies	Deleted
Class 3 - Combination, self-contained	Class 3
Type II. Flaring, impact-actuated	Type II
Class 1 - Tube coupling	Class 1
Class 2 - Hand block	Class 2
Style A - Single flare	Deleted
Style B - Double flare	Style B
Class 3 - Vise blocks	Class 3
Class 4 - Hinged blocks	Class 4
Class 5 - Removable dies	Deleted
Type III. Reading, rotary	Deleted
Class 1 - Bench	Deleted
Class 2 - Portable	Deleted

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MILITARY CUSTODIANS:

Army - GL
Navy - SH
Air Force - 84

Review activities:

Army - GL, AV
Navy - SH
Air Force - 84, 85

User activities:

Army - AT, ME, CE
Navy - MC, YD

Military coordinating activity:

Army - GL

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price of this specification 16 cents each.

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